

Jordan E. Bisanz

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Employment

Pennsylvania State University **State College PA, USA**
Assistant Professor of Biochemistry and Molecular Biology 2021-
Dorothy Foehr and J. Lloyd Huck Early Career Chair in Host-Microbiome Interactions

University of California San Francisco **San Francisco CA, USA**
Postdoctoral Scholar and Fellow 2015-2021
Principal Investigator: Peter Turnbaugh

Training and Education

Doctor of Philosophy **London ON, Canada**
Department of Microbiology and Immunology, University of Western Ontario 2010-2015
Graduate Advisor: Gregor Reid

Institut Pasteur de Lille **Lille, France**
Visiting Student 2015
Collaborators: Bruno Pot and Benoit Foligné

National Institute of Medical Research **Mwanza, Tanzania**
Visiting Student 2012
Collaborators: Joseph Mwanga and John Chagalucha

Bachelor of Medical Sciences **London ON, Canada**
Schulich School of Medicine and Dentistry, University of Western Ontario 2006-2010
Honors Specialization in Microbiology and Immunology with Distinction
Western Scholars Honors Society

Extramural Research Funding, Fellowships and Awards

Total extramural funding at PSU (PI/MPI): 5.3 million
Total extramural funding at PSU (all roles): 14.9 million

Award for Early Career Applied and Biotechnological Research 2026
American Society for Microbiology

Clarivate Highly Cited Researchers 2025
Recognized in the Cross-Field category based on citation performance across publications from the preceding 11-year period.

R35 GM151045 (PI: Bisanz)
National Institute of General Medical Sciences 2023-2028
Project title: Decoding Microbial Diversity in the Human Gut Microbiome.

R01 ES038206 (MPIs: Tian, Bisanz, Peters)
National Institute of Environmental Health Sciences 2026-2030
Project Title: Environmental PFAS alter microbial function, impair host metabolism, and drive disease progression.

R00 AI147165 (PI: Bisanz)

National Institute for Allergy and Infectious Diseases

2021-2023

Project title: Diet-microbe interactions modulating host energy balance.

Young Investigator Grant in Probiotics Research (PI: Bisanz)

Global Probiotics Council / Danone and Yakult

2023

Project title: In with the good, out with the bad: engineering the microbiome to modulate dietary metal absorption.

R35 ES035027 (PI: Patterson, Co-I Bisanz)

National Institute of Environmental Health Sciences

2023-2031

Project title: Environmental Chemical Impact on the Host-Microbiome Interaction

R01 DK133334 (PI: Singh, Co-I Bisanz)

National Institute of Diabetes and Digestive and Kidney Diseases

2023-2028

Project title: Interplay of processed diet, gut microbiota, and interferon-linked mucosal immunity in the onset and prevalence of inflammatory bowel disease.

K99 AI147165 (PI: Bisanz)

National Institute for Allergy and Infectious Diseases

2020-2022

Project title: Diet-microbe interactions modulating host energy balance.

Application received impact score of 10.

NSERC Postdoctoral Fellowship

Natural Sciences and Engineering Research Council of Canada

2015-2017

180 awarded internationally, ranked in top 15% of applications

Alexander Graham Bell Doctoral Canada Graduate Scholarship

Natural Sciences and Engineering Research Council of Canada

2012-2015

261 awarded nationally, ranked in top 15% of applications

Ontario Graduate Scholarship

Ontario Ministry of Training, Colleges and Universities, declined 2013

2011-2013

Students for Development Grant

Canadian International Development Agency

2012

Publications**Total publications: 66****Total as a corresponding author: 11****Total as a first author: 12****Total manuscripts with PSU affiliation: 30****Citations: 36,315* H-index: 39* I-10 index: 53***

*Data collected from Google Scholar 11 June 2026

Underlined individuals represent direct trainees at PSU

1. Mulisa G, J Zhao, G Lelissa, IJ Ademola-Popoola, A Diaz Huemme, LS Weyrich, A Mihret, T Gemechu, A Bane, R Pero-Gascon, S De Saeger, T Abebe and **JE Bisanz**. 2026. Multi-cohort comparative analysis of

salivary microbiotas reveals rural Ethiopians harbor a distinct composition correlated with lower esophageal cancer prevalence. **mSystems**. e0023226. doi: 10.1128/msystems.00232-26.

2. Boyle N, JJ Sekela, M Wang, H Mannocho-Russo, JJ Pyo, MS Kim, S Tian, I Koo, ES Friedman, C Tanes, M Anitha, Y Tian, E Morgan, IA Murray, JP Zackular, K Bittinger, JD Lewis, GH Perdew, GD Wu, BS Zemel, PC Dorrestein, **JE Bisanz**, MR Redinbo and AD Patterson. **2026**. Glucuronidation Metabolomic Fingerprinting to Map Host-Microbe Metabolism. **Nature Communications**. doi: 10.1038/s41467-026-73398-1.
3. Babdor J, RK Patel, B Davidson, C Noecker, M Rahim, **JE Bisanz**, I Tenvooren, D Marquez, M Calvo, V Johri, E MacCarthy, A Shaheed, C Ekstrand, AM Weakley, FB Yu, O Fiehn, PJ Turnbaugh, AJ Combes, GK Fragiadakis and MH Spitzer. **2026**. Coordinated variation in the immune systems and microbiomes of healthy humans is linked to tonic interferon states. **Cell**. 189:3144-3163.e20.
4. Foley L, S Kohli, S Tian, L Eisaman, H Ganga, G Hamner, M Aronson, **JE Bisanz** and SH Medina. **2026**. Core-shell commensal biocapsules for in situ gut microbiome engineering. **Bioactive Materials**. 65:796-808.
5. Morgan EW, AJ Annalora, DM Coslo, K Gowda, D Desai, F Dong, EW Davis, IA Murray, F Hao, I Koo, KS Petersen, PM Kris-Etherton, T Wolfe, R Erickson, S Walk, **JE Bisanz**, SG Amin, CB Marcus, AD Patterson and GH Perdew. **2026**. Interkingdom glycine conjugates of indole-3-carboxylates are Ah receptor ligands. **Communications Biology**. *In Press*. Preprint doi: 10.64898/2026.02.13.705761.
6. Liu P, MS Nitchman, K Jump, S Tian, R Mani, **JE Bisanz** and P Kalantari. **2026**. A STING agonist protects against exacerbation of schistosome egg-induced immunopathology. **PLoS Pathogens**. 22(7):e1014394.
7. Alsaadi A, L Welz, AA Pothakamury, MS Yahia, C Gilloteau, P Prasad, J Sadler, JD Smith, BC Jenkins, GS Diven, J Bornhäuser, T Mekdoud, S Tian, P Rosenstiel, S Schreiber, **JE Bisanz**, K Aden and MR McReynolds. **2026**. Tracing NAD⁺ metabolism uncovers adaptive coordination between host and microbiome during colitis. **Cell Reports**. 45(8):117730.
8. Bywater A, A Novaski Seffrin, **JE Bisanz**, F Di Gioia and J Kovac. **2026**. Soilless system design impacts the diversity and composition of microbiota. **Applied and Environmental Microbiology**. *In Press*.
9. Tian S, MS Kim, J Zhao, K Heber, F Hao, D Koslicki, S Tian, V Singh, AD Patterson and **JE Bisanz**. **2025**. A designed synthetic microbiota provides insight to community function in *Clostridioides difficile* resistance. **Cell Host & Microbe**. 33(3):373-387.
10. Heber K, S Tian, D Betancurt-Anzola, H Koo and **JE Bisanz**. **2025**. StrainR2 accurately deconvolutes strain-level abundances in synthetic microbial communities. **Bioinformatics**. 41(8):btaf440.
11. Koo H, K Heber, S Tian, ST Connolly, F Hao, J Zhao, B Swencki-Underwood, AD Patterson, GE Townsend and **JE Bisanz**. **2025**. A synthetic gut microbiota provides understanding of the maintenance and functional impact of phage. **mBio**. e0234125:10.1128/mbio.02341-25.
12. Kim MS and **JE Bisanz**. **2025**. Design and application of synthetic human gut microbial communities. **Gut Microbes**. 17(1):2575923.
13. Li X, L Xu, B Demaree, C Noecker, **JE Bisanz**, DW Weisgerber, C Modavi, PJ Turnbaugh and A Abate. **2025**. Microbiome single cell atlases generated with a commercial instrument. **Advanced Science**. e2409338.

14. DiMattia ZS, J Zhao, F Hao, S Koshkin, **JE Bisanz**, AD Patterson, JA Fleming, PM Kris-Etherton and KS Petersen. **2025**. Effect of varying quantities of lean beef as part of a mediterranean-style dietary pattern on gut microbiota and plasma, fecal, and urinary metabolites: a randomized crossover controlled feeding trial. **Journal of the American Heart Association**. 14(9):e041063.
15. Riley TM, T Wright, R Lamendella, **JE Bisanz**, JC See, K Kanani, P Kris-Etherton and K Petersen. **2025**. Nighttime pistachio consumption alters stool microbiota diversity and feature abundance compared to education to consume 1-2 carbohydrate exchanges (15-30g) over 12 weeks in adults with prediabetes: a secondary analysis from a randomized cross over trial. **Current Developments in Nutrition**. 9(7):107481.
16. JM Kelliher, C Mirzayi, SR Bordenstein, A Oliver, CA Kellogg, EL Hatcher, M Berg, P Baldrian, M Aljumaah, CML Miller, C Mungall, V Novak, A Palucki, E Smith, N Tabassum, G Bonito, JR Brister, PSG Chain, M Chen, S Degregori, JP Dundore-Arias, JB Emerson, VMC Fernandes, R Flores, A Gonzalez, ZA Hansen, SA Jackson, AMM Moustafa, TR Northen, N Pariente, J Pett-Ridge, S Record, L Reji, A-L Reysenbach, VI Rich, L Richardson, S Roux, LM Schriml, RS Shabman, MA Sierra, MB Sullivan, P Sundaramurthy, KM Thibault, LR Thompson, S Tighe, E Vereen, **STREAMS Consortium** and EA Elloe-Fadrosch. **2025**. STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies. **Nature Microbiology**. 10(12):3059-3068.
17. Mulisa G, R Pero-Gascon, V McCormack, **JE Bisanz**, FR Talukdar, T Abebe, M De Boevre and S De Saeger. **2025**. Multiple mycotoxin exposure assessment through human biomonitoring in an esophageal cancer case-control study in the Arsi-Bale districts of Oromia region of Ethiopia. **International Journal of Hygiene and Environmental Health**. 362:114466.
18. Kyaw TS, M Sandy, K Trepka, JJN Goh, K Yu, V Dimassa, EN Bess, **JE Bisanz*** and PJ Turnbaugh*. **2024**. Human gut Actinobacteria boost drug absorption by secreting P-glycoprotein ATPase inhibitors. **iScience**. 27(6):110122. ***Corresponding Authors**
19. Anderson BD, DE Sepulveda, R Nachnani, A Cortez, A Beckett, **JE Bisanz***, JJ Kellogg* and WM Raup-Konsavage*. **2024**. High cannabigerol hemp extract moderates colitis and modulates the microbiome in an inflammatory bowel disease model. **Journal of Pharmacology and Experimental Therapeutics**. 390(3):331-341. ***Corresponding Authors**
20. McCurry MD, GD D'Agostino, JT Walsh, **JE Bisanz**, I Zalosnik, J Korzenik, X Dong, DJ Morris, JR Korzenik, AG Edlow, EP Balskus, PJ Turnbaugh, JR Huh and AS Devlin. **2024**. Gut bacteria convert glucocorticoids into progestins in the presence of hydrogen gas. **Cell**. 187(12):2952-2968.
21. Zhao J, YW Sun, KM Chebn, C Aliaga, **JE Bisanz** and K El-Bayoumy. **2024**. Black raspberry modulates cecal and oral microbiome at the early stage of a dibenzo[def,p]chrysene-induced murine oral cancer model. **Cancer Prevention Research**. 18(1):11-21.
22. Tian Y, B Rimal, **JE Bisanz**, W Gui, TM Wolfe, I Koo, IM Murray, SK Nettleford, S Yokoyama, F Dong, KS Prabhu, PJ Turnbaugh, ST Walk, GH Perdew, and AD Patterson. **2024**. Early-life exposure to a potent aryl hydrocarbon receptor ligand results in persistent changes to the microbiota and host glucose homeostasis. **Environmental Health Perspectives**. 132(8):087005.
23. Sepulveda DE, KE Vrana, JJ Kellogg, **JE Bisanz**, D Desai, NM Grazaine and WM Raup-Konsavage. **2024**. The Potential of Cannabichromene (CBC) as a Therapeutic Agent. **Journal of Pharmacology and Experimental Therapeutics**. 391(2):206-213.

24. Rimal B, SL Collins, CE Tanes, ER Rocha, MA Granda, S Solanki, NJ Hoque, EC Gentry, I Koo, ER Reilly, F Hao, D Paudel, V Singh, T Yan, MS Kim, K Bittinger, JP Zackular, KW Krausz, D Desai, S Amin, JP Coleman, YM Shah, **JE Bisanz**, FJ Gonzalez, JP Vanden Heuvel, GD Wu, PC Dorrestein, EE Weinert and AD Patterson. **2024**. Bile Salt Hydrolase Catalyses Formation of Amine-Conjugated Bile Acids. **Nature**. 626(8000):859-863.
25. Carmody RC and **JE Bisanz**. **2023**. Roles of the gut microbiome in weight management. **Nature Reviews Microbiology**. 21(8):535-550.
26. Kim MS and **JE Bisanz**. **2023**. Meta-organismal Metabolism in Cardiovascular Disease. **Cell Host & Microbe**. 31(1):1-2.
27. Anderson BD and **JE Bisanz**. **2023**. Challenges and Opportunities of Strain Diversity in Gut Microbiome Research. **Frontiers in Microbiology**. 14:1117122.
28. Upadhyay V, R Suryawanshi, P Tasoff, M McCavitt-Malvido, G Kumar, V Murray, C Noecker, **JE Bisanz**, Y Hswen, C Ha, B Sreekumar, I Chen, S Lynch, M Ott, S Lee and PJ Turnbaugh. **2023**. Mild SARS-CoV-2 infection results in long-lasting microbiota instability. **mBio**. 14(4):e00889-23.
29. Noecker C, J Sanchez, **JE Bisanz**, V Escalante, M Alexander, K Trepka, A Heinken, Y Liu, D Dodd, I Thiele, B DeFelice and PJ Turnbaugh. **2023**. Systems biology elucidates the distinctive metabolic niche filled by the human gut microbe *Eggerthella lenta*. **PLoS Biology**. 21(5):e3002125.
30. Tian S and **JE Bisanz**. **2022**. Making Gut Microbiomes from Scratch. **Cell Host & Microbe**. 30(11):1508-1509.
31. Collins SL, JG Stine, **JE Bisanz**, CD Okafor and AD Patterson. **2022**. Bile acids and the gut microbiome: metabolic interactions and impacts on disease. **Nature Reviews Microbiology**. 21(4):236-247.
32. Paik D, L Yao, Y Zhang, S Bae, GD D'Agostino, E Kim, EA Franzosa, J Avila-Pacheco, **JE Bisanz**, CK Rakowski, H Vlamakis, RJ Xavier, PJ Turnbaugh, RS Longman, MR Krout, CB Clish, C Huttenhower, JR Huh and AS Sloan. **2022**. Human gut bacteria produce T_H17-modulating bile acid metabolites. **Nature**. 603(7903):907-912.
33. Sbierski-Kind J, S Grenkowitz, S Schlickeiser, A Sandforth, M Friedrich, D Kunkel, R Glauben, S Brachs, K Mai, A Thürmer, A Radonić, O Drechsel, PJ Turnbaugh, **JE Bisanz**, HD Volk and R Jumpertz von Schwartzberg. **2022**. Effects of caloric restriction on the gut microbiome are linked with immune senescence. **Microbiome**. 10(1):57.
34. Jumpertz von Schwartzberg R*, **Bisanz JE***, S Lyalina, P Spanogiannopoulos, QY Ang, J Cai, S Dickman, M Friedrich, S.-Y. Liu, SL Collins, D Ingebrigsten, S Miller, JA Turnbaugh, AD Patterson, KS Pollard, K Mai, J Spranger and PJ Turnbaugh. **2021**. Caloric restriction disrupts the microbiota and colonization resistance. **Nature**. 595:272-277. **Equal contributions*.
35. Lam KN, P Spanogiannopoulos, P Soto-Perez, A Alexander, MJ Nalley, **JE Bisanz**, FB Yu and PJ Turnbaugh. **2021**. Phage-delivered CRISPR-Cas9 for strain-specific depletion and genomic deletions in the gut microbiota. **Cell Reports**. 37(5):109930.
36. Ang QY, DL Alba, V Upadhyay, **JE Bisanz**, J Cai, HL Lee, E Barajas, G Wei, AD Patterson, SK Koliwad and PJ Turnbaugh. **2021**. Difference in the gut microbiomes of distinct ethnicities within the same geographic area are linked to host metabolic health. **eLife**. e70349.

37. C Mirzayi, A Renson, Genomic Standards Consortium, Massive Analysis and Quality Control Society, F Zohra, S Elsafoury, L Geistlinger, LJ Kasselmann, K Eckenrode, J van de Wijgert, A Loughman, FZ Marques, DA MacIntyre, M Arumugam, R Azhar, F Beghini, K Bergstrom, A Bhatt, **JE Bisanz**, J Braun, H Corrada Bravo, GA Buck, F Bushman, D Casero, G Clarke, M Carmen Collado, PD Cotter, JF Cryan, RT Demmer, S Devkota, E Elinav, JS Escobar, J Fettweis, RD Finn, AA Fodor, S Forslund, A Franke, C Furlanello, J Gilbert, E Grice, B Haibe-Kains, S Handley, P Herd, S Holmes, JP Jacobs, L Karstens, R Knight, D Knights, O Koren, DS Kwon, M Langille, B Lindsay, D McGovern, AC McHardy, S McWeeney, NT Mueller, L Nezi, M Olm, N Palm, E Pasolli, J Raes, MR Redinbo, M Rühlemann, R Balfour Sartor, PD Schloss, L Schriml, E Segal, M Shardell, T Sharpton, E Smirnova, H Sokol, JL Sonnenburg, S Srinivasan, LB Thingholm, PJ Turnbaugh, V Upadhyay, RL Walls, P Wilmes, T Yamada, G Zeller, M Zhang, N Zhao, L Zhao, W Bao, A Culhane, V Devanarayan, J Dopazo, X Fan, M Fischer, W Jones, R Kusko, CE Mason, TR Mercer, SA Sansone, A Scherer, L Shi, S Thakkar, W Tong, R Wolfinger, C Hunter, N Segata, C Huttenhower, JB Dowd, HE Jones and L Waldron. **2021**. Reporting guidelines for human microbiome research: the STORMS checklist. **Nature Medicine**. 27:1885-1892.
38. **Bisanz JE**, P Soto Perez, C Noecker, AA Aksenov, KN Lam, EN Bess, HJ Haiser, TS Kyaw, FB Yu, VM Rekda, CWY Ha, S Devkota, EP Balskus, PC Dorrestein, E Allen-Vercoe and PJ Turnbaugh. **2020**. A genomic toolkit for the mechanistic dissection of intractable human gut bacteria. **Cell Host & Microbe**. 27(6):1001-1013.
39. Bess EN, **JE Bisanz**, F Yarza, A Bustion, B Rich, X Li, S Kitamura, E Waligurski, QY Ang, D Alba, P Spanogiannopoulos, S Nayfach, S Koliwad, D Wolan, A Franke and PJ Turnbaugh. **2020**. Genetic basis for the cooperative bioactivation of plant lignans by *Eggerthella lenta* and other human gut bacteria. **Nature Microbiology**. 5(1):56-66.
40. Man K, V Escalante, C Bowman, Y Tian, **JE Bisanz**, J Nikkanen, K Ganeshan, B Wang, AD Patterson, PJ Turnbaugh and A Chawla. **2020**. A thermogenic fat-epithelium axis regulates intestinal tissue tolerance. **PNAS**. 117(50):32029-32037.
41. Greenstein RA, RR Barrales, N Sanchez, **JE Bisanz**, S Braun and B Al-Sady. **2020**. Set1/COMPASS repels heterochromatin invasion at euchromatic sites by disrupting Suv39/Clr4 activity and nucleosome stability. **Genes and Development**. 34(1-2):99-117.
42. Rekda VM, Bernardino PN, Luescher MU, Kiamehr S, Le C, **JE Bisanz**, PJ Turnbaugh, EN Bess and EP Balskus. **2020**. A widely distributed metalloenzyme class enables gut microbial metabolism of host- and diet-derived catechols. **eLife**. 9:e50845.
43. Pröbstel AK, L Schirmer, X Zhou, K Kim, R Baumann, A Ramesh, AL Greenfield, R Dandekar, S Vistnes, RD Schubert, OL Rojas, S Singh, EB Tran, P Barba, **JE Bisanz**, PJ Turnbaugh, LR Shio, J Graves, BAC Cree, SL Hauser, JL Gommerman, MR Wilson and SE Baranzini. **2020**. Gut microbiota-specific B cells traffic to the brain in active multiple sclerosis. **Science Immunology**. 5(53):eabc7191.
44. **Bisanz JE**, V Upadhyay, JA Turnbaugh and PJ Turnbaugh. **2019**. Meta-analysis reveals reproducible gut microbiome alterations in response to a high-fat diet. **Cell Host & Microbe**. 26(2):265-272.
45. Soto Perez P*, **JE Bisanz***, JD Berry, KN Lam, J Bondy-Denomy and PJ Turnbaugh. **2019**. CRISPR-Cas system of a prevalent human gut bacterium reveals hyper-targeting against phages in a human virome catalog. **Cell Host & Microbe**. 26(3):325-335. ***Equal contributions**.

46. Carmody RN, **JE Bisanz**, CF Maurice, S Lyalina, BP Bowen, KB Louie, D Treen, K Chaideh, VM Rekdal, EN Bess, P Spanogiannopoulos, KC Bauer, TW Balon, KS Pollard, TR Northen and PJ Turnbaugh. **2019**. Cooking shapes the structure and function of the gut microbiome. **Nature Microbiology**. 4(12):2052-2063.
47. Rekdal VM, EN Bess, **JE Bisanz**, PJ Turnbaugh and EP Balskus. **2019**. Discovery and inhibition of an interspecies gut bacterial pathway for Levodopa metabolism. **Science**. 364(6445):eaau6323.
48. Chagwedera DN, QY Ang, **JE Bisanz**, K Ganeshan, J Cai, AD Patterson, PJ Turnbaugh and A Chawla. **2019**. Nutrient sensing in CD11c cells alters the gut microbiome to regulate food intake and body mass. **Cell Metabolism**. 30(2):364-373.
49. E Bolyen, JR Rideout, MR Dillon, NA Bokulich, CC Abnet, GA Al-Ghalith, H Alexander, EJ Alm, M Arumugam, F Asnicar, Y Bai, **JE Bisanz**, K Bittinger, A Brejnrod, CJ Brislawn, CT Brown, BJ Callahan, AM Caraballo-Rodríguez, J Chase, EK Cope, R Da Silva, C Diener, PC Dorrestein, GM Douglas, DM Durall, C Duvallet, CF Edwardson, M Ernst, M Estaki, J Fouquier, JM Gauglitz, SM Gibbons, DL Gibson, A Gonzalez, K Gorlick, J Guo, B Hillmann, S Holmes, H Holste, C Huttenhower, GA Huttley, S Janssen, AK Jarmusch, L Jiang, BD Kaehler, KB Kang, CR Keefe, P Keim, ST Kelley, D Knights, I Koester, T Kosciolk, J Kreps, MGI Langille, J Lee, R Ley, YX Liu, E Loftfield, C Lozupone, M Maher, C Martoz, BD Martin, D McDonald, LJ McIver, AV Melnik, JL Metcalf, SC Morgan, JT Morton, AT Naimey, JA Navas-Molina, LF Nothias, SB Orchanian, T Pearson, SL Peoples, D Petras, ML Preuss, E Priesse, LB Rasmussen, A Rivers, MS Robeson, P Rosenthal, N Segata, M Shaffer, A Shiffer, R Sinha, SJ Song, JR Spear, AD Swafford, LR Thompson, PJ Torres, P Trinh, A Tripathi, PJ Turnbaugh, S Ul-Hasan, JJJ van der Hooft, F Vargas, Y Vásquez-Baeza, E Vogtmann, M von Hippel, W Walters, Y Wan, M Wang, J Warren, KC Weber, CHD Williamson, AD Willis, ZZ Xu, JR Zaneveld, Y Zhang, Q Zhu, R Knight and JG Caporaso. **2019**. Reproducible, interactive, scalable and extensive microbiome data science using QIIME 2. **Nature Biotechnology**. 37(8):852-857.
50. **Bisanz JE**, P Spanogiannopoulos, LM Pieper, AE Bustion and PJ Turnbaugh. **2018**. How to determine the role of the microbiome in drug disposition. **Drug Metabolism and Disposition**. 46(11):1588-1596.
51. Koppel N, **JE Bisanz**, ME Pandelia, PJ Turnbaugh and EP Balskus. **2018**. Discovery and characterization of a prevalent human gut bacterial enzyme sufficient for the inactivation of a family of plant toxins. **eLife**. 7:e33953.
52. Al KF, **JE Bisanz**, GB Gloor, G Reid and JP Burton. **2018**. Evaluation of sampling and storage procedures on preserving the community structure of stool microbiota: a simple at-home toilet-paper collection method. **Journal of Microbiological Methods**. 144:117-121.
53. Daisley BA, M Monachese, M Trinder, **JE Bisanz**, JA Chmiel, JP Burton and G Reid. **2019**. Immobilization of cadmium and lead by *Lactobacillus rhamnosus* GR-1 mitigates apical-to-basolateral heavy metal translocation in a Caco-2 model of the intestinal epithelium. **Gut Microbes**. 10(3):321-333.
54. Sinha R, G Abu-Ali, E Vogtmann, AA Fodor, B Ren, A Amir, E Schwager, J Crabtree, S Ma, **The Microbiome Quality Control Project Consortium**, CC Abnet, R Knight, O White and C Huttenhower. **2017**. Assessment of variation in microbial community amplicon sequencing by the Microbiome Quality Control (MBQC) project consortium. **Nature Biotechnology**. 35(11):1077-1086.
55. Harvie RM, AW Chisholm, **JE Bisanz**, JP Burton, P Herbison, K Schultz and M Schultz. **2017**. Long-term irritable bowel syndrome symptom control with reintroduction of selected FODMAPs. **World Journal of Gastroenterology**. 23(25):4632-4643.

56. **Bisanz JE**, P Suppiah, M Thomson, T Milne, N Yeoh, A Nolan, G Ettinger, G Reid, GB Gloor, JP Burton, M Cullinan and S Stebbings. **2016**. The oral microbiome of patients with axial spondyloarthritis compared to healthy individuals. **PeerJ**. 4:e2095.
57. **Bisanz JE**, MK Enos, G PrayGod, S Seney, JM Macklaim, S Chilton, D Willner, R Knight, C Fusch, G Fusch, GB Gloor, JP Burton and G Reid. **2015**. The microbiota at multiple body sites during pregnancy in a rural Tanzanian population and the effects of Moringa supplemented probiotic yogurt. **Applied and Environmental Microbiology**. 81(15):4965-4975.
58. Trinder M, **JE Bisanz**, JP Burton and G Reid. **2015**. Probiotic lactobacilli: a potential prophylactic treatment for reducing pesticide absorption in humans and wildlife. **Beneficial Microbes**. 6(6):841-847.
59. McMillan A, S Rulisa, M Sumarah, JM Macklaim, J Renaud, **JE Bisanz**, GB Gloor and G Reid. **2015**. A multi-platform metabolomics approach identifies highly specific biomarkers of bacterial diversity in the vagina of pregnant and non-pregnant women. **Scientific Reports**. 5:14174.
60. **Bisanz JE**, MK Enos, J Mwanga, J Chagalucha, JP Burton, GB Gloor and G Reid. **2014**. Randomized open-label pilot study of the influence of probiotics and the gut microbiome on toxic metal levels in Tanzanian pregnant women and school children. **mBio**. 5(5):e01580-14.
61. **Bisanz JE**, JM Macklaim, GB Gloor and G Reid. **2014**. Bacterial metatranscriptome analysis of a probiotic yogurt using an RNA-Seq approach. **International Dairy Journal**. 39(2):284-292.
62. **Bisanz JE**, S Seney, A McMillan, R Vongsa, D Koenig, GB Gloor, M Sumarah, B Ford, D Herman, JP Burton and G Reid. **2014**. A systems biology approach investigating the effect of *Lactobacillus rhamnosus* GR-1 and *L. reuteri* RC-14 on the vaginal microbiome and host responses in a double blind, placebo-controlled clinical trial of postmenopausal women with intermediate Nugent scores. **PLoS ONE**. 9(8):e104511.
63. Reid JN, **JE Bisanz**, M Monachese, JP Burton and G Reid. **2013**. The rationale for probiotics improving reproductive health and pregnancy outcomes. **American Journal of Reproductive Immunology**. 69(6):558-566.
64. Hummelen R*, JM Macklaim*, **JE Bisanz***, JA Hammond, A McMillan, R Vongsa, D Koenig, GB Gloor and G Reid. **2011**. Vaginal microbiome and epithelial gene array in post-menopausal women with moderate to severe dryness. **PLoS ONE**. 6(11):e26602. ***Equal contributions**.
65. **Bisanz JE** and G Reid. **2011**. Unraveling how probiotic yogurt works. **Science Translational Medicine**. 3(106):106ps41
66. Macphee RA, R Hummelen, **JE Bisanz**, WL Miller and G Reid. **2010**. Probiotic strategies for the treatment and prevention of bacterial vaginosis. **Expert Opinion on Pharmacotherapy**. 11(18):2985-2995.

Patents

Bisanz JE, S Tian, E Weinert, B Underwood. A synthetic microbiota and related processes and apparatuses. US Patent Application Filed 13 February 2026: PCT/US26/15233.

Patterson AD, Peters JM, **Bisanz JE**, Tian Y. Bioaccumulation of persistent environmental chemicals by bacteria. US Provisional Patent Application Filed 16 February 2026: 63/955,777.

Bisanz JE, G Reid, M Monachese, JET Van Hylckama Vlieg, T Smokvina, JP Burton. Food grade bacteria for the removal of toxic compounds. EU Patent #EP2833898B1, US Patent # US10487305, Canada Patent #CA2869566 , Chinese Patent #CN104755090B.

Bisanz JE, G Reid, M Monachese, JET Van Hylckama Vlieg, T Smokvina, JP Burton. *Lactobacillus rhamnosus* CNCM I-4716 food grade bacteria. US Patent #US9540609B2, EU Patent #EP3031931B1.

Oral Presentations (including confirmed future presentations)

American Association of Biological Anthropologists Annual Meeting	Philadelphia PA, USA – 2027
PennCHOP Microbiome Symposium	Philadelphia PA, USA – 2026
Microbe 2026 (American Society for Microbiology)	Washington DC, USA – 2026
Anaerobe 2026 (Anaerobe Society of the Americas)	New York NY, USA – 2026
American Society of Nutrition (ASN) 2026	Washington DC, USA – 2026
American Chemical Society Fall Meeting	Chicago IL, USA – 2026
Columbia University Department of Systems Biology Seminar	New York NY, USA – 2026
Benioff Center for Microbiome Medicine Symposium (UCSF)	San Francisco CA, USA – 2026
Microbes in Microbiomes (Microbiology Society)	Miami FL, USA – 2026
American Chemical Society Fall Meeting	Washington DC, USA – 2025
Ohio State University Department of Microbiology	Columbus OH, USA – 2025
University of Utah Department of Pathology Seminar	Salt Lake UT, USA – 2024
Trends Microbiology Metagenomic Approaches in Microbiome Research	Online Webinar – 2024
American Chemical Society World Microbiome Day	Online Webinar – 2024
NCCIH Workshop on Endocrine-Disrupting Chemicals	Bethesda MD, USA – 2024
Qiagen Human Microbiome Webinar Series	Online Webinar – 2024
AdventHealth Translational Research Institute	Orlando FL, USA – 2024
Microbiome Data Congress	Boston MA, USA – 2023
Leibniz Institute for Natural Product Research and Infection Biology	Jena, Germany – 2023
Claflin University Biology Department	Orangeburg, SC, USA – 2023
U Penn Big Data in Biomedical and Population Health Sciences	Philadelphia PA, USA – 2022
American Association of Pharmaceutical Scientists PharmSci 2022	Boston MA, USA – 2022
Microbiome and Probiotic R&D Forum	Miami FL, USA – 2022
International Conference on the Bioscience of Lipids	Online Webinar – 2022
Penn State Nutritional Sciences Colloquium	State College PA, USA – 2022
Penn State Department of Food Science Seminar	State College PA, USA – 2022
Penn State Microbiome Center Seminar Series	State College PA, USA – 2021
ETH Zurich Institute of Food, Nutrition and Health Seminar (online)	Zurich, Switzerland – 2021
UC Berkeley CCB/SPH Bioinformatics Seminar	Berkeley CA, USA – 2020
PSU Biochemistry and Molecular Biology Seminar	State College PA, USA – 2020
University of Illinois UC Microbial Systems Initiative Seminar	Urbana IL, USA – 2020
UC Irvine Molecular Biology and Biochemistry Seminar	Irvine CA, USA – 2020
University of Washington Microbiology Seminar	Seattle WA, USA – 2020
Chan Zuckerberg Microbiome Initiative Symposium	San Francisco CA, USA – 2020
Dartmouth Microbiology and Immunology Seminar	Lebanon NH, USA – 2019
UC Davis Innovation Institute for Food & Health Summit	Davis CA, USA – 2019
Infectious Disease Association of California Winter Symposium	Oakland CA, USA – 2019
UCSF Diabetes and Obesity Retreat	Santa Cruz CA, USA – 2018
Anaerobe Society of the Americas: Anaerobe 2018	Las Vegas NV, USA – 2018
UCSF Microbial Pathogenesis and Host Defense Seminar Series	San Francisco CA, USA – 2017
UCSF Research in Progress Seminars	San Francisco CA, USA – 2017
Charité Hospital Center for Cardiovascular Research Seminar	Berlin, Germany – 2016
Berchtesgaden Microbiome Science Days	Berchtesgaden, Germany – 2016
Beneficial Microbes Conference	The Hague, Netherlands – 2015
Institut Pasteur Seminar Series	Lille, France – 2015
International Union of Microbiological Societies	Montreal QC, Canada – 2014
International Scientific Association for Probiotics and Prebiotics	Aberdeen, Scotland – 2014
Exploring Human Host-Microbiome Interactions in Health and Disease	Cambridge, England – 2013

Curriculum Vitae

Jordan E Bisanz

University of Reading: Whiteknights Guest Seminar
National Institute for Medical Research Guest Seminar
Canadian Society of Microbiologists Student Symposium Competition

Reading, England – 2013
Mwanza, Tanzania – 2012
St. John's NL, Canada – 2011

Poster Presentations

Keystone Microbiome Symposium
International Scientific Association for Probiotics and Prebiotics
Probiotics, Prebiotics and the Host Microbiome
Lawson Health Research Day
Lawson Health Research Day
Microbes for Health 2nd International Symposium
International Human Microbiome Congress

Banff AB, Canada – 2018
Washington DC, USA – 2015
New York NY, USA – 2013
London ON, Canada – 2013
London ON, Canada – 2011
Paris, France – 2011
Vancouver BC, Canada – 2011

Trainees/Staff

Name	Timeframe	Position after lab	Awards
Graduate trainees:			
Susan Tian	2021 – 2025	Postdoc at UCSF, Susan Lynch	- Paul Berg Prize for Outstanding Student in Molecular Biology at PSU
Ben Anderson	2021 – present		- Stephen B. Brumbach Distinguished Graduate Fellowship
Min Soo Kim	2022 – present		- NIAID F31 AI183815
Shane Connolly	2023 – present		- NIDDK T32 DK120509
Eric Antonian	2025 – present		- Stephen B. Brumbach Distinguished Graduate Fellowship - NIDDK T32 DK120509 - NIGMS T32 GM156692 - NIGMS T32 GM156692
Postdocs:			
Jingcheng Zhao	2022 – present		- National Dairy Council Early Career Investigator Award
Research staff:			
Jaden Carper	2021 – 2022	Biology Teacher, Shikellamy School District	
Wei Gui	2023 – 2023	Associate Product Manager, Abbott Labs	
Bethany Swencki-Underwood	2024 – present		
Undergraduate researchers:			
Kaitlyn Do	2025 – present		- CHOP CRISP internship
Shayan Gupta	2025 – present		- Johnson and Johnson oncology internship
Dorcas Falokun	2024 – present		- Millennium Scholars Program
Joy Rao	2026 – present		- NASA/PSU URIP internship
Heejung Koo	2021 – 2025	PhD Candidate, UPenn Medical Science	- ECoS Undergraduate Research Award - ABRCMS travel award
Dean Miller	2021 – 2023	PhD Candidate, Princeton Chemical and Biological Engineering	- ECoS Undergraduate Research Award - John and Paige Lapinski Undergraduate Fellowship
Ahmed Al-Thenyan	2021 – 2022	Captain, Alpha Fire Company	
Lilyana Hasan	2023 – 2023	Microbiologist, PA Department of Health	
Nelson Dinham	2023 – 2023	PSU Undergrad	
Isabella Neuman	2023 – 2023	PSU Undergrad	
Anastacia Diaz Huemme	2023 – 2024	Research Lab Tech, UPitt	
Garrick Zheng	2023 – 2024	Medical Student, PSU COM	
Lexie Bialek	2023 – 2024	Medical Assistant, Dermatology Partners	
Kerim Heber	2023 – 2025	PhD Candidate, UPenn Genomics and Comp Bio	- NSF GRFP
Aviauna Beckett	2023 – 2025	Medical Lab Scientist program, Thomas Jefferson University	
Calyn Bainey	2025 – 2025	Medical Student, LECOM	

Teaching Experience

Instructor (MICRB/VBSC410 Principles of Immunology) **2023-present**

Department of Biochemistry and Molecular Biology / Veterinary and Biomedical Sciences, PSU

Taught course employing active learning methodologies to cover major topics relevant to immunology.

Developed course material on mucosal immunology, systems immunology, and interpretation of experimental data. Students range from 2nd year undergraduates to graduate students. Annual spring enrollment: 71-119.

Instructor (BMB 502 Critical Scientific Analysis) **2024, 2026**

Co-taught course focusing on critical thinking in the analysis of scientific manuscripts, poster presentations, and seminars. Enrollment: 17 graduate students.

Instructor (BMB 521 Special Topics in Microbiology) **2024, 2026**

Coordinated team-taught course focusing on recent developments in microbiology for graduate students. My sessions focused on metagenomics, statistics, and data literacy. Enrollment: 8-10 graduate students.

Instructor (Microbial Bioinformatics Workshop Series) **2020**

Benioff Center for Microbiome Medicine

Prepared and co-taught sessions on BASH, package managers, cluster computing, QIIME2, R, and the Tidyverse. Developed in response to need for training opportunities during COVID-19 shelter-in-place order.

Guest Instructor (BMS270/BMI219 The Human Microbiome) **2017-2021**

Biomedical Science / Biological and Medical Informatics Programs (UC San Francisco)

Delivered lecture and tutorial for manipulating high-throughput DNA sequencing and exploratory data analysis of 16S rRNA amplicon sequencing data. Instructor evaluations: 4.5/5 (2018), 4.8/5 (2019).

Tutorial content: jbisanz.github.io/BMS270_BMI219

Lecturer/Teaching Assistant (Microbiology for Nursing 3820A) **2013-2014**

Department of Microbiology and Immunology, Western University

Delivered 1-hour lectures on prevention, diagnosis, and treatment of *Neisseria*, *Chlamydia*, and *Spirochetes* for 3rd year undergraduate course of 302 students. Prepared, administered, and graded randomized case studies comprising 25% of final grade.

Teaching Assistant (Introduction to Biology 1002B) **2014**

Department of Biology, Western University

Delivered 2-hour lectures and facilitated student discussion groups covering topics relating to scientific literacy and research.

Teaching Assistant (Genetics 2581A) **2013**

Department of Biology, Western University

Delivered 1.5-hour tutorial lectures on basic concepts in prokaryotic and eukaryotic genetics.

Teaching Assistant (Biology of Prokaryotes Lab Course) **2011**

Department of Microbiology and Immunology, Western University

Taught basic bacteriology lab skills including media preparation, culture, differential biochemical identification tests and API-20E identification.

Internal Service

Penn State Microbiome Center Executive Committee	2023 – Present
Faculty Advisor: Penn State Gnotobiotics Facility	2021 – Present
Faculty Director: One Health Microbiome Center Co-laboratory	2022 – Present
Co-director: Center for Molecular Toxicology and Carcinogenesis	2025 – Present
Member: Penn State Cancer Institute Shared Resource Scientific Advisory Committee	2026 – Present
Microbiology Faculty Search Committee Department of Biochemistry and Molecular Biology, PSU	2024 – 2026
Teaching Faculty Search Committee Department of Biochemistry and Molecular Biology, PSU	2025 – 2026
Chair and member: Microbiology Undergraduate Curriculum Committee Department of Biochemistry and Molecular Biology, PSU	2024 – Present
Toxicology Faculty Search Committee Department of Veterinary and Biomedical Sciences/Institute for Energy and the Environment, PSU	2024 – 2025
Department Head Search Committee Department of Biochemistry and Molecular Biology, PSU	2022 – 2023
Graduate Recruitment Committee Biochemistry, Microbiology, and Molecular Biology PhD Program, PSU	2021 – 2023, 2026 – Present
Climate and Diversity Committee Department of Biochemistry and Molecular Biology, PSU	2021 – 2022

External Service

Standing Member: Interspecies Microbial Interactions and Infections (IMII) National Institutes of Health – Term beginning Fall 2026	2026 – 2030
Board of Editors: mBio American Society for Microbiology	2026 – 2029
Associate Editor: Anaerobe Anaerobe Society of the Americas/Elsevier	2021 – Present
Analytical Microbiology Steering Committee for MSACL 2026 Association for Mass Spectrometry and Advances in the Clinical Lab	2025 – 2026
Ad hoc reviewer: Etiology, Diagnostic, Intervention Treatment of Infectious Diseases (EDIT) National Institutes of Health	2026
Ad hoc reviewer: Member Conflict – Topics in Endocrinology, Metabolism and Reproductive Systems (EMS) National Institutes of Health	2025
Ad hoc reviewer: Interspecies Microbial Interactions and Infections Study Section (IMII) National Institutes of Health	2025
Ad hoc reviewer: Member Conflict - Applied Immunology and Disease Control National Institute of General Medical Sciences	2025
Ad hoc reviewer: Directorate for Technology, Innovation and Partnership National Science Foundation	2025
Ad hoc reviewer: Directorate for Biological Sciences National Science Foundation	2025

Curriculum Vitae

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Ad hoc reviewer: Genes, Cells and Molecules Section Committee

2024 – 2025

Natural Sciences and Engineering Research Council of Canada

Ad hoc reviewer: Drug and Biological Disposition and Toxicity Study Section (DBDT)

2023 – 2024

National Institutes of Health

Ad hoc reviewer: Collaborative Program Grant for Multidisciplinary Teams (RM1)

2024

National Institute of General Medical Sciences

Ad hoc Referee

2020 – Present

Flanders Research Foundation (FWO) Postdoctoral Fellowship Committee, Graduate Women in Science Fellowship Committee, Benioff Microbiome Center Grant Reviewer, Austrian Science Fund, Polish National Science Center, Wellcome Trust, Wellcome LEAP

Ad hoc Peer Reviewer

2015 - present

218 completed assignments from 60 journals. Selected examples with number of assignments in parentheses: Cell (5), Cell Host & Microbe (10), Cell Metabolism (2), Nature Microbiology (17), Nature Medicine (3), Nature Metabolism (3), Nature Immunology (3), Science Translational Medicine (3), Gut Microbes (11), mBio (10), mSystems (7), Microbiome (28), Genome Biology (5), Genome Medicine (5)

Outreach

Matters Microbial Podcast

2026

Discussing how synthetic communities are key to understanding the function of complex microbial communities in the human gut and beyond.

Science, Microbes & Health Podcast

2023

International Scientific Association for Probiotics and Prebiotics

Interviewed to discuss the bacterial species concept and how it applies to microbiome research and probiotic applications. Available at: isappscience.org/episode-31-microbial-species/.

tvN (South Korea)

2023

Interviewed to discuss the human microbiome in metabolic health as part of the weekly TV show “The Power of Superfoods” shown on the major cable network tvN.

Science Outreach Panelist

2023

State Theatre (State College PA, USA)

Part of an expert panel at a showing of the documentary “The Invisible Extinction” to answer audience questions about the human microbiome.

Guest Speaker

2022

American Society for Microbiology Student Chapter

Delivered talk to PSU undergraduates on the microbiome research.

Guest Speaker

2019

NutrigenomX/ Global Society for Genetics and Genome Biology (University of California Berkeley)

Delivered talk to UCB undergraduate students on the microbiota in health and its interactions with diet.

Science Outreach Panelist

2014

London Public Library (London ON, Canada)

Part of an expert panel at a screening of the documentary Microbirth to answer audience questions about the human microbiome and probiotics.

Students and Fellows Association Executive**2013-2015**

International Scientific Association for Probiotics and Prebiotics

Co-organized conferences (Aberdeen 2014, Washington DC 2015) and led recruitment efforts for new members.

Relevant Additional Training**Mentoring Matters Workshop****2026**

Penn State Office of the Senior Vice President for Research

Half-day professional development focusing on strategies for effective mentorship of graduate and undergraduate trainees.

Evidence Based Teaching Academy**2025**

Eberly College of Science

A 1-week professional development program aiding instructors to redesign and refine their courses using research-supported teaching practices. Combined workshops, peer discussion, and hands-on course planning.

Responsible Conduct of Research Course**2021**

UCSF Office for Postdoctoral Scholars

Over 8 one-hour sessions, this course addresses important issues including societal implications of scientific misconduct, entrepreneurship, collaborative science, responsible authorship, animal welfare, implications of genomic research, mentorship, and racism in science.

Inclusive Research Mentorship Course**2020**

UCSF Office of Career and Professional Development

A 12 hour course covering topics relating to hiring, training, and supervising team members from diverse backgrounds.

Scientific Leadership and Management Skills Course**2019**

Gladstone Institutes

Practical strategies and theories for beginning and leading research groups in an academic environment. 16-hours covering the following topics: motivating, managing conflict, strategic time and resource management, staffing, building lab culture, establishing a research program, vision and mission statements, and negotiation.

Relevant Work Experience**London Public Health Laboratory****2008-2009**

Seasonal Laboratory Assistant

Ontario Ministry of Health/ Ontario Public Health Agency

Supervisors: Abdul Chagla, and Christine Fry.